

IEEE Club Event Management System

Detailed Workflow & Architecture Overview

1. Event Administration & Governance

- Admin creates event: Title, Fees, Start/End Dates, Capacity.
- All events are queued for Super Admin Approval.
- Approved events are published to the public list automatically.

2. Registration & Intelligent Gating

- Users/Guests register with validation (Uniqueness check per event).
- Fee-based events move to "Pending Payment".
- Free events move directly to "Approved" with Ticket generation.

3. Secure Payment Lifecycle

- Participants upload proof of payment via mobile/desktop.
- Finance Admins review proofs in a dedicated terminal.
- System transitions Reference IDs (TEMP to KARE) upon verification.

4. Real-time Attendance (QR)

- Each approved participant receives a permanent QR Code Ticket.
- Admins use the "Scan & Mark" tool at venue checkpoints.
- Verification ensures correct event, status, and single-use logic.

5. Certificate Issuance & Cloud Storage

- Admins design high-res certificate templates visually.
- Batch generation processes only "Attended & Approved" users.
- PDFs are securely synced to Google Drive and Supabase Storage.

This document provides a summary of the system logic as of March 2026.